

End-Term Semester Examination 2024

Paper set: 1

Name of the program: **B. Tech**

Semester: **7th**

Name of the course: **Pharmaceutical Biotechnology**

Course code: **TBT 703a**

Time: 3hrs

Maximum Marks: 100

- Q. 1: a) Highlight important characteristics of drugs and their physicochemical properties (CO-1)
b) Define a drug and its different dosage forms
c) Implement the principle of pharmacodynamics in drug action
- Q. 2: a) Sketch out different dosage forms of drug and their routes of administration (CO-2)
b) Analyze different drug receptor forms and pharmacokinetics
c) Implement the drug distribution process and their bioavailability in the system
- Q. 3: a) Implement good manufacturing practices in drug development (CO-3)
b) Create a plan for the production of monoclonal antibodies using hybridoma technology
c) Defend the usage of biomaterials in the medical field and their usage
- Q. 5: a) Justify proteins as therapeutic molecules and their routes of administration (CO-4)
b) Categorize the drug delivery systems and their applications in the modern medical field
c) Appraise bioreactors and their applications in biopharmaceutical field
- Q. 4: a) Justify the production of monoclonal antibodies and their usage in the treatment of various health disorders (CO-5)
b) Correlate important factors that are common among cytokines, interferons, interleukins, and tumour necrosis factors
c) Analyse methods for different drug-delivery approaches and therapeutic proteins

